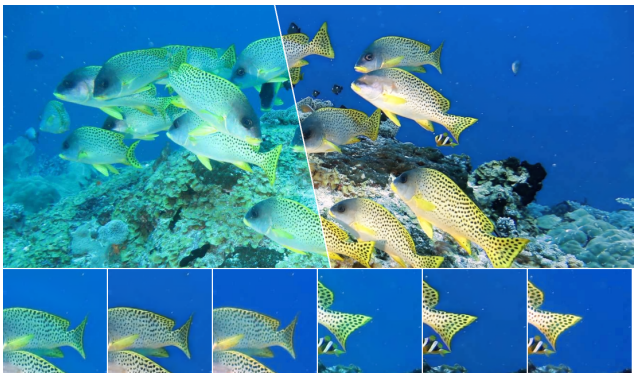
# SandBAR: A Band-Asymmetric Architecture for Underwater Image Restoration with Removable Frequency Scaffolding

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## Abstract

*Reliable underwater imagery supports marine monitoring, ecological assessment and autonomous exploration, yet light scattering, absorption and turbidity jointly degrade image clarity, distort colour and suppress fine structure. Although these effects are physically interwoven, they commonly appear as complementary frequency tendencies: large-scale, channel-dependent appearance distortions and local high-frequency detail loss. We refer to this tendency as band asymmetry; it motivates different forms of restoration for global colour and local structure. Our primary contribution is SandBAR, a novel unified band-asymmetric architecture for underwater image restoration. Rather than treating frequency processing as an isolated module, SandBAR organises the restoration pipeline around band asymmetry at the block, encoder and objective levels. Within this architecture, we introduce a novel band-asymmetric sandwich block with a reserved post-attention repair slot and the Split-band Parallel Asymmetric Repair Kernel (SPARK), a novel module that applies channel-aware large-support correction to the low band, which predominantly carries large-scale appearance distortions, and lightweight local refinement to the detail-bearing high band, thereby restoring global colour while preserving fine structure. To deliver the global cast to the depth at which the network represents global structure, a learnable low-frequency guidance path injects a low-pass view of the input into the deep encoder, and a three-scale spatial-frequency objective supervises the same axis throughout the decoder. Building on the reserved post-attention slot, we further introduce Post-attention Scaffold Training (PaST), a novel removable nonlinear scaffolding scheme that trains the permanent SandBAR graph under band-specific capacity and runs it at inference without that capacity. SandBAR reaches 29.89 / 28.92 / 22.77 dB on EUVP / LSUI / UIEB and establishes a new three-domain mean state of the art on the benchmark family, exceeding the strongest published result by approximately 1.0 dB. PaST adds no parameters or FLOPs at inference and preserves matched latency.*



(a) UIR-PolyKernel (b) SandBAR (c) Ground Truth (d) UIR-PolyKernel (e) SandBAR (f) Ground Truth

Figure 1. **Visual comparison of SandBAR and a competing method.** Top: input and SandBAR. SandBAR restores global colour while preserving texture detail.

## 1. Introduction

Underwater image degradation is not a single uniform loss of visual quality. Light scattering, absorption and turbidity jointly reduce visibility, distort colour and blur fine structure. Their effects are physically interwoven: attenuation and backscatter vary with water conditions, scene range and content, so no single mechanism belongs exclusively to one spatial-frequency band [1, 2]. In the image domain, however, the resulting degradation often exhibits complementary frequency tendencies. Large-scale, channel-dependent distortions dominate background colour and smooth ap-

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pearance, whereas edges, textures and small objects concentrate high-frequency structure and lose contrast first. We refer to this empirical tendency as **band asymmetry**. It motivates smooth, channel-structured correction for the low-frequency component and local detail refinement for the high-frequency component. Figure 1 illustrates this behaviour: SandBAR corrects the large-scale colour cast while preserving fine object texture and boundaries.

Deep underwater restoration has advanced rapidly, from physics-guided CNNs [16, 17] through transformers [24, 28] to recent spatial–spectral state-space models [23]. Band asymmetry nevertheless enters these architectures in one of two ways. Uniform backbones leave it implicit and rely on a homogeneous stack to discover the distinction during training. Frequency-aware designs [6, 14] model it explicitly through decomposition and specialised per-band processing. These approaches generally treat band asymmetry either implicitly or within an individual component, rather than organising block topology, encoder guidance and supervision around it as one underwater restoration architecture. Moreover, explicit frequency operations remain active at inference, so their additional branches and band-specific transformations must be executed for every restored image. This creates both an architectural gap and a practical tension between explicit frequency modelling and efficient inference.

These limitations motivate our primary goal: to build a unified underwater restoration architecture that makes band asymmetry explicit throughout the restoration pipeline. A further challenge is whether the nonlinear band-specific capacity used by such an architecture must remain in the network at inference. Content-adaptive decomposition, asymmetric per-band processing and gated recombination are nonlinear operations, so they cannot be algebraically folded into neighbouring layers in the same way as linear reparameterisation branches can [7, 8].

Our main contribution is SandBAR, a novel unified band-asymmetric architecture for underwater image restoration. It translates this tendency into three coordinated levels of design. At the block level, a sandwich transformer block gives every stage two feed-forward slots—one before attention for generic mixing, one after attention reserved for band-specific processing—where a standard restoration block [28] has one. At the encoder level, a learnable low-frequency guidance path delivers the global cast to the depth at which the network represents global structure. At the objective level, three decoder heads are supervised in both the spatial and the frequency domain. At its core is the Split-band Parallel Asymmetric Repair Kernel (SPARK), a novel module that occupies the reserved slot: it applies channel-aware large-support correction to the low band, which predominantly carries large-scale appearance distortions, and lightweight local refinement to the detail-bearing high band, thereby restoring global colour while preserving surviving detail.

To make this architecture inference-efficient, we further introduce Post-attention Scaffold Training (PaST), a novel removable nonlinear scaffolding scheme within SandBAR. SPARK modulates a residual around the slot input, so removing it collapses the block onto its scaffold-free form and the single constant it leaves behind folds into the adjacent normalisation. Figure 3 summarises the resulting trade-off. Relative to the separately trained scaffold-free permanent graph, PaST raises the three-domain mean PSNR from 26.79 to 27.19 dB at the same 24.09M-parameter inference size. Retaining SPARK at inference increases the graph to 28.59M parameters and reaches 27.20 dB, only 0.0036 dB above the post-fold result using the unrounded values. SandBAR reaches 29.89 / 28.92 / 22.77 dB on EUVP / LSUI / UIEB and attains a three-domain mean of 27.19 dB, exceeding the strongest published result on the benchmark family by approximately 1.0 dB. PaST obtains this with no added parameters or FLOPs at inference and at matched latency.

Our contributions are:

- **Our primary contribution is SandBAR, a novel unified band-asymmetric architecture for underwater image restoration.** It makes band asymmetry an explicit architectural prior and instantiates it at the block, encoder and objective levels: sandwich blocks reserving a post-attention slot for band repair, a learnable low-frequency guidance path into the deep encoder, and a three-scale spatial-frequency objective. The three levels operate as one system organised around a single axis.
- **Within SandBAR, we introduce a novel band-asymmetric sandwich block and the novel SPARK split-band repair module.** The block reserves a post-attention slot for band repair, while SPARK treats the two frequency bands with operators matched to their dominant content.
- **We further introduce PaST, a novel removable nonlinear scaffolding scheme for training SandBAR.** Band-specific capacity shapes the network during joint training, after which the scaffold is removed and the permanent SandBAR graph runs inference without it.
- **SandBAR establishes a new three-domain mean state of the art on the benchmark family**, exceeding the strongest published result by approximately 1.0 dB and obtaining the PaST benefit with no added parameters or FLOPs at inference and at matched latency.

## 2. Related work

**Underwater image restoration.** Early learning-based methods paired physical priors with CNNs: WaterNet [16] fuses pre-processed inputs under a gated net-

work, Ucolor [17] embeds a medium-transmission prior into a multi-colour-space encoder, and FUnIE-GAN [13] targets real-time enhancement adversarially. Transformer architectures raised accuracy at higher cost: the U-shape Transformer [24] introduced the LSUI benchmark, and general restoration backbones such as Restormer [28] transfer strongly to the task. Recent work explores semi-supervised learning [12] and spatial–spectral state-space models [23]. Across this line, underwater-specific capacity generally remains part of the permanent inference graph. SandBAR instead makes band asymmetry an architecture-wide organising principle across block topology, encoder guidance and supervision. Within this architecture, task-specific frequency capacity shapes the network during training without becoming permanent inference cost.

**Frequency-aware restoration.** Decomposing features by frequency is an established design in general restoration: SFNet [6] selects informative bands with content-adaptive filters, and frequency-domain transformers [14] compute attention spectrally. SandBAR instantiates its band split with SFNet's content-adaptive low-pass filter [6] and its gated recombination. Within the overall SandBAR architecture, SPARK advances frequency-aware restoration by assigning different operators to the low- and high-frequency bands according to their dominant content rather than treating them symmetrically. PaST further makes the complete in-block frequency path training-only rather than permanent.

**Training-time-only capacity.** Three families discard capacity between training and inference. Structural re-parameterisation—RepVGG [8], Diverse Branch Blocks [7], edge-oriented blocks for super-resolution [30]—trains multi-branch *linear* topologies that are algebraically collapsed into single convolutions. Deep supervision [15] discards auxiliary heads attached *outside* the backbone. Distillation [11] discards an entire teacher network but requires one to exist. Unlike linear structural re-parameterisation, external auxiliary heads or teacher–student distillation, PaST is the training mechanism within SandBAR that trains a single network under removable *nonlinear* in-block scaffolding and, after one joint optimisation run, retains only its permanent graph at inference.

## 3. SandBAR architecture

Figure 2 presents the complete SandBAR architecture and shows how its constituent designs form one coordinated band-asymmetric restoration system. SandBAR translates band asymmetry into three levels of design: a post-attention slot for asymmetric band repair (Sec. 3.1–3.2), a low-frequency guidance path for global colour correction (Sec. 3.4), and a three-scale spatial-frequency objective (Sec. 3.5). SPARK occupies the reserved slot during training, while PaST removes its frequency machinery before inference (Sec. 3.3).

### 3.1. Band-asymmetric sandwich block

To give band repair a dedicated position after global context mixing, every SandBAR stage uses a sandwich block with feed-forward slots before and after attention:

$$\mathbf{x}_1 = \mathbf{x}_0 + \mathrm{G}_{\mathrm{pre}}(\mathrm{LN}(\mathbf{x}_0)), \tag{1}$$

$$\mathbf{x}_2 = \mathbf{x}_1 + \mathrm{A}(\mathrm{LN}(\mathbf{x}_1)), \tag{2}$$

$$\mathbf{x}_3 = \mathbf{x}_2 + \mathrm{S}(\mathrm{LN}(\mathbf{x}_2)), \tag{3}$$

where A is multi-Dconv-head transposed attention [28], $\mathrm{G}_{\mathrm{pre}}$ is a gated-Dconv FFN (GDFN) [28], and S is the post-attention slot. A standard restoration block has a single feed-forward network; reserving a second slot separates band repair from generic channel mixing and places it where global context has just been aggregated. In the inference graph the slot holds a second GDFN; during training it holds SPARK. Both slots sit on residual branches, which is what makes the slot's content removable (Sec. 3.3). This sandwich topology forms the block-level core of SandBAR.

### 3.2. SPARK: asymmetric split-band repair

To restore global colour without suppressing surviving detail, SPARK applies different operators to the low- and high-frequency components of each post-attention feature. Given the slot input $\mathbf{z} = \mathrm{LN}(\mathbf{x}_2)$, a content-adaptive low-pass filter [6] predicts a per-image, softmax-normalised $K \times K$ kernel and produces

$$\mathbf{z}_l = \mathcal{F}_{\mathrm{lp}}(\mathbf{z}), \qquad \mathbf{z}_h = \mathbf{z} - \mathbf{z}_l, \tag{4}$$

an exact two-band partition. The bands then receive asymmetric treatment:

$$\mathbf{u}_l = \mathrm{PW}(\mathrm{DW}_{5\times5}(\mathrm{SE}(\mathbf{z}_l))), \quad \mathbf{u}_h = \mathrm{PW}(\mathrm{DW}_{3\times3}(\mathbf{z}_h)), \tag{5}$$

where SE is squeeze–excitation channel attention, $\mathrm{DW}_k$ a depthwise convolution and PW a pointwise convolution. The low band receives channel attention and large support to address the large-scale, channel-structured colour and illumination variations that predominantly appear there. The detail-bearing high band receives a small kernel and no channel-global reweighting, which could suppress already weakened fine structure. A gated recombination [6] weighs the bands per channel,

$$\begin{aligned} \mathbf{v} &= \mathrm{PW}_{\mathrm{out}}(\alpha_l \odot \mathbf{u}_l + \alpha_h \odot \mathbf{u}_h), \\ [\alpha_l, \alpha_h] &= \sigma\big(\mathrm{FC}(\mathrm{GAP}(\mathbf{u}_l + \mathbf{u}_h))\big), \end{aligned} \tag{6}$$

with $\sigma$ a softmax over the concatenated per-channel attention logits of the two branches, and the module closes residually through a GDFN:

$$\mathrm{S}(\mathbf{z}) \;=\; \mathrm{G}_{\mathrm{post}}(\mathbf{v} + \mathbf{z}). \tag{7}$$

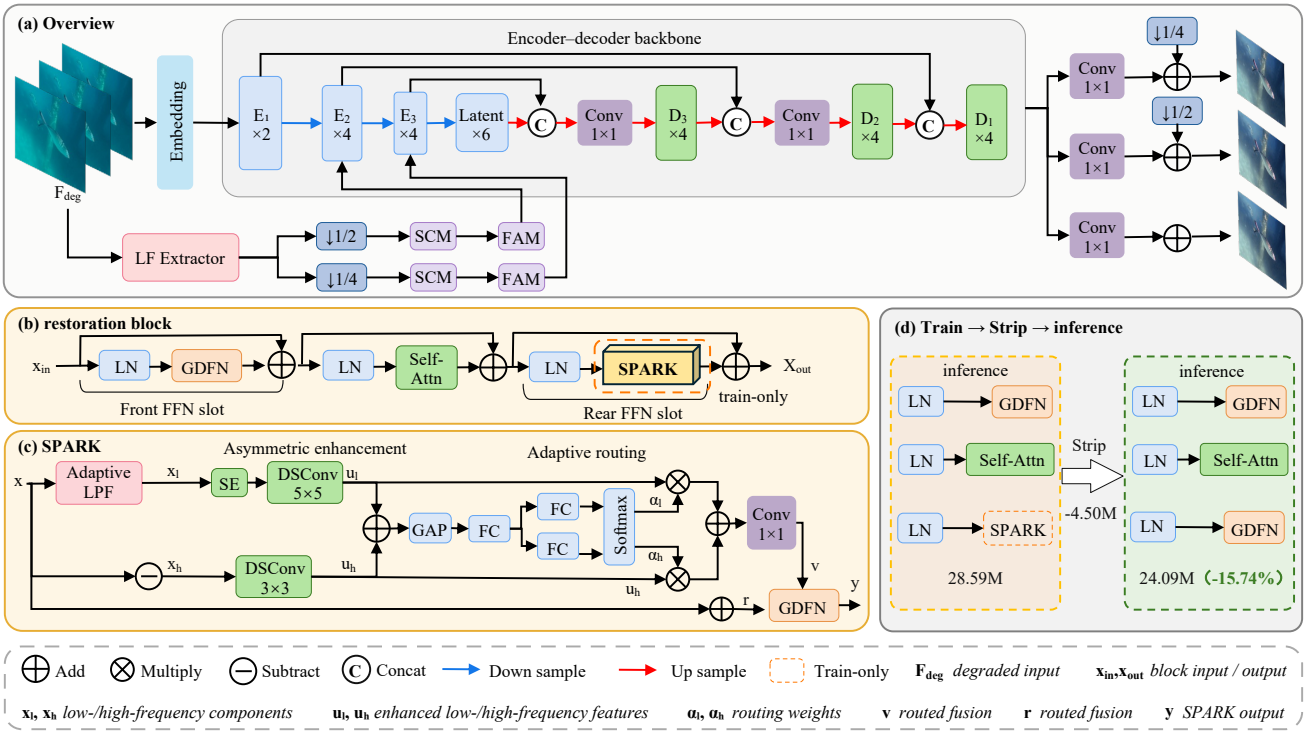



Figure 2. **SandBAR architecture and its constituent designs.** **(a)** Encoder–decoder backbone with a learnable low-frequency guidance path injected at encoder levels 2 and 3, and three supervised output heads. **(b)** The sandwich block: two feed-forward slots around attention, with SPARK occupying only the post-attention slot. **(c)** SPARK: content-adaptive band split, asymmetric per-band repair, gated recombination and residual closing. **(d)** PaST: after joint training the scaffold is removed and its bias folded, and what remains at inference is the permanent scaffold-free block.

### 3.3. PaST: removable nonlinear frequency scaffolding

Building on the reserved post-attention slot, PaST turns SPARK into removable nonlinear training scaffolding. Equations (4)–(6) are nonlinear, so SPARK admits no absorbing algebra and cannot be collapsed into neighbouring layers as a linear branch can [8]. PaST makes removability architectural instead. Because the entire apparatus modulates a *residual* around the slot input (Eq. (7)), deleting it collapses $\mathrm{S}(\mathbf{z})$ onto $\mathrm{G}_{\mathrm{post}}(\mathbf{z})$—the permanent scaffold-free block—up to the constant bias $\mathbf{b}_{\mathrm{out}}$ of $\mathrm{PW}_{\mathrm{out}}$, which folds into the affine bias of the preceding LayerNorm:

$$\mathbf{b}'_{\mathrm{LN}} \;=\; \mathbf{b}_{\mathrm{LN}} + \mathbf{b}_{\mathrm{out}}. \tag{8}$$

The bias fold is algebraically exact and numerically indistinguishable in floating-point execution (Supplementary). The inference network is then, layer for layer and parameter for parameter, SandBAR's permanent graph: 28,586,687 training parameters become 24,088,479, and 146.86 GFLOPs at $256^2$ become 132.15. PaST therefore trains the permanent SandBAR graph under richer band-specific capacity while leaving none of the in-block frequency machinery at inference. It is the training mechanism that makes SandBAR's nonlinear band-repair scaffold removable.

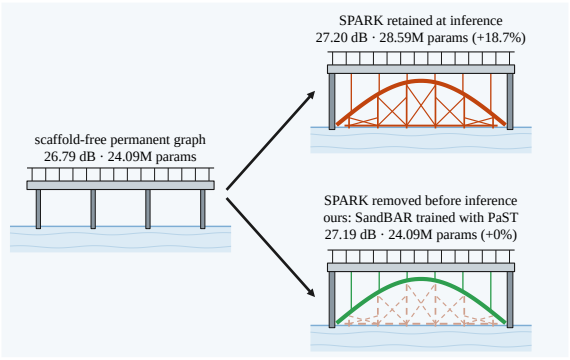



Figure 3. **PaST preserves performance while reducing parameters and inference cost.** Removing SPARK before inference restores the permanent graph's parameter count, FLOPs and latency.

### 3.4. Low-frequency guidance into the encoder

To deliver the global colour cast directly to encoder stages that represent global structure, SandBAR injects a learnable low-pass view of the input at deeper resolutions. The encoder levels that carry global structure meet the input

only through cascaded downsampling, by which point the cast is entangled with content. A $31 \times 31$ depthwise Gaussian, initialised at $\sigma = 10$ and trained end to end, extracts $\mathbf{I}_{\text{lf}}$; downsampled copies at $1/2$ and $1/4$ are encoded by shallow convolutional modules and merged into encoder levels 2 and 3 by feature-attention fusion [4, 6]. The path therefore supplies the low-band problem where the network can act on it globally, complementing the block-level design at the encoder level.

### 3.5. Three-scale spatial-frequency objective

To align supervision with the same band-asymmetric prior, SandBAR optimises three decoder outputs in both the spatial and the frequency domain. Decoder levels 3, 2 and 1 each carry an output head producing a residual prediction at $1/4$, $1/2$ and full resolution. With $\hat{\mathbf{y}}_s$ the prediction and $\mathbf{y}_s$ the reference at scale $s$,

$$\mathcal{L} = \sum_s \left\|\hat{\mathbf{y}}_s - \mathbf{y}_s\right\|_1 + \lambda \sum_s \left\|\mathcal{T}(\hat{\mathbf{y}}_s) - \mathcal{T}(\mathbf{y}_s)\right\|_1, \quad (9)$$

where $\mathcal{T}$ stacks the real and imaginary parts of the 2D FFT and $\lambda = 0.1$, following the multi-scale spatial-plus-frequency supervision common in restoration [4, 6]. The objective therefore supervises global appearance and local structure across the complete decoder hierarchy and completes SandBAR's architecture-wide coordination at the objective level.

## 4. Experiments

### 4.1. Data and evaluation

We follow the public 4,800-pair underwater restoration benchmark [10] (800 UIEB [16] + 2,000 EUVP [13] + 2,000 LSUI [24]) and evaluate on its corresponding 490-image paired test set, comprising 200 EUVP, 200 LSUI and 90 UIEB images. SandBAR uses an encoder–decoder with dim 32, blocks $(2, 4, 4, 6)$, heads $[1, 2, 4, 8]$ and expansion 2.66. Final models are trained for 650 epochs with Adam on random $256 \times 256$ crops using a batch size of 4. The learning rate is warmed up for three epochs and then cosine-annealed from $10^{-4}$ to $10^{-6}$. Controlled architectural ablations in Tables 4 and 5 use the same training configuration with a 100-epoch budget. Complete training settings are provided in the supplementary material. Cost figures are measured on one RTX 5090; the complete timing procedure is also provided in the supplementary material.

### 4.2. State-of-the-art restoration

As a complete system, SandBAR establishes the strongest three-domain mean PSNR on the benchmark family, reaching 27.19 dB and exceeding the strongest published result by approximately 1.0 dB (Table 1). It reaches 29.89 dB on EUVP and 28.92 dB on LSUI. MAC-Lookup [27] obtains a higher LSUI SSIM of 0.931 versus our 0.922, using explicit SSIM supervision together with stacked multi-stage local/global refinement. The resulting system is comparatively compute-heavy at 173.2 GMACs, and this structural-similarity advantage is accompanied by a clear reconstruction-fidelity trade-off: its LSUI PSNR is 27.78 dB, 1.14 dB below SandBAR's 28.92 dB. SandBAR therefore delivers substantially higher pixel-level fidelity while retaining competitive structural similarity.

### 4.3. Zero-cost inference through PaST

We next isolate PaST, the removable-scaffolding contribution within SandBAR, from the complete architecture.

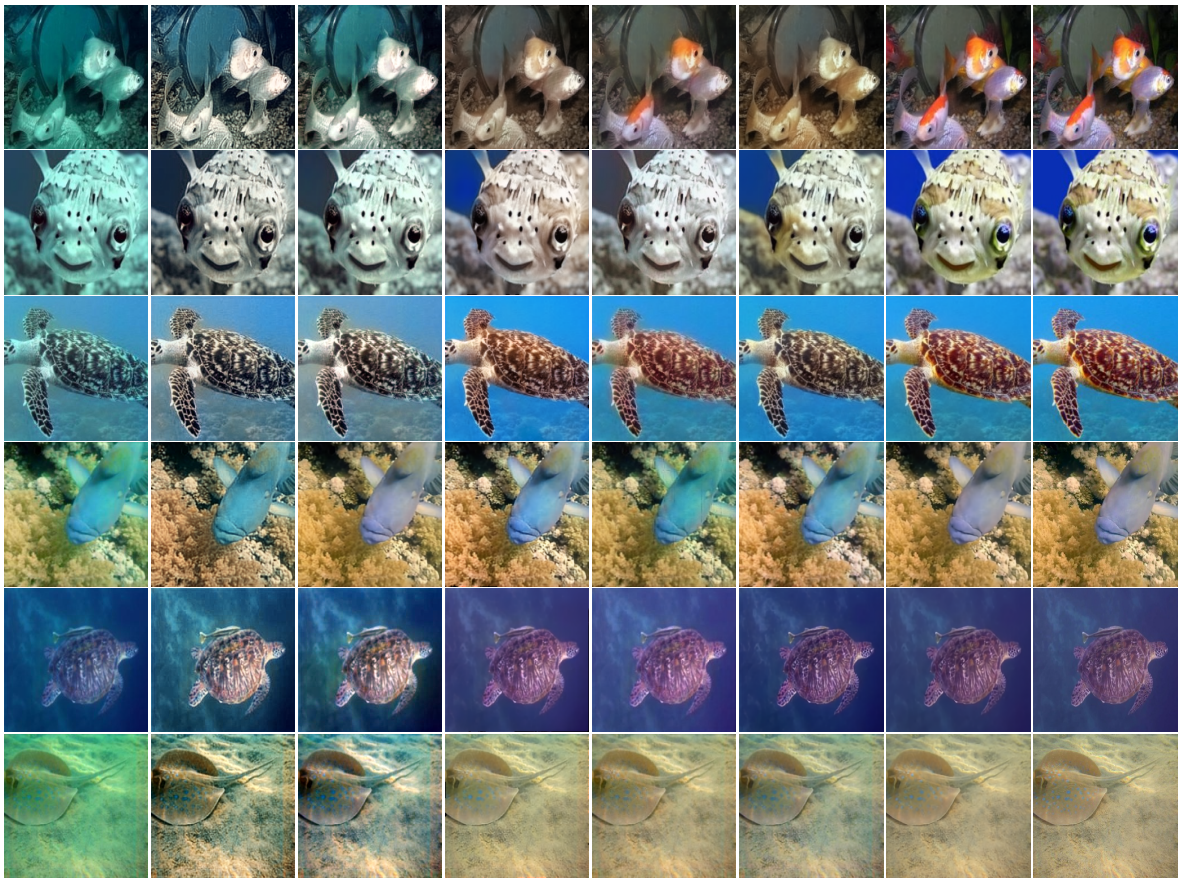
PaST places the complete cost of SPARK in training. Removing the scaffold reduces the graph from 28,586,687 to 24,088,479 parameters and from 146.86 to 132.15 GFLOPs, while the strip-and-fold PaST model matches the permanent SandBAR graph in latency at both resolutions (Table 2). The first row evaluates the trained SandBAR model before strip-and-fold, with SPARK retained at inference; it is not the graph used for the reported SandBAR inference results. Retaining SPARK gives a three-domain mean PSNR of 27.20 dB, whereas the post-fold inference graph gives 27.19 dB. The unrounded values differ by only 0.0036 dB, showing that strip-and-fold preserves restoration performance while removing the frequency scaffold from the inference graph.

Figure 5 makes the two effects visually distinct. Strip-and-fold changes the trained model's output negligibly, whereas training with the scaffold changes the permanent graph substantially: the post-fold–plain mean absolute difference is $20.1\times$, $53.8\times$ and $139.2\times$ larger on EUVP, LSUI and UIEB, respectively, than the retained–post-fold difference. Despite identical inference cost, PaST improves PSNR on all three domains: by 0.39, 0.40 and 0.43 dB on EUVP, LSUI and UIEB, a three-domain mean of 0.41 dB. These gains are computed from the underlying unrounded scores of the last two rows of Table 1. **No part of the frequency scaffold is present in the post-fold inference graph.** These results demonstrate that removable nonlinear frequency capacity strengthens the permanent restorer without entering its inference graph. Figure 6 visualises the split the scaffold learns before it is removed: the low branch concentrates the colour cast, the high branch the fading detail.

With the same permanent inference architecture, SandBAR exceeds the unsplit variant by 0.126 dB PSNR and 0.00249 SSIM.

### 4.4. Where band repair belongs

The preceding experiment establishes that band-specific capacity can shape a stronger permanent graph without remaining at inference. We next ask why that capacity is assigned specifically to the post-attention slot. Table 4 isolates placement from module and backbone identity. With



(a) Input (b) Semi-UIR (c) URSCT (d) UIR-PolyKernel (e) U-Transformer (f) MAC-Lookup (g) Ours (h) Reference

Figure 4. Samples for visual comparison showcase how SandBAR achieves more faithful colour restoration and clearer structural detail by repairing the low- and high-frequency bands with separate operators, while the compared baselines either retain the input's cast or soften surviving detail.

SPARK in SandBAR, moving band repair from the pre-attention slot to the post-attention slot raises the three-domain mean from 26.43 to 26.72 dB. Replacing SPARK with DFFN [14] reproduces the effect, from 26.25 to 26.52 dB. Changing the backbone instead gives the same result: SPARK in Restormer [28] improves from 26.15 to 26.44 dB when moved after attention. Across all three controlled comparisons, post-attention placement contributes 0.27 to 0.29 dB. The gain therefore follows the slot across two different modules and two different backbones.

The result exposes a functional asymmetry between the two feed-forward positions. The pre-attention FFN operates before global context has been assembled, and any frequency-specific transformation it produces is subsequently re-mixed by attention. The post-attention FFN receives the context-aware representation instead and can translate the global view directly into restoration. For underwater degradation, this ordering is particularly important: scene-level evidence establishes the global colour, haze and illumination state first, after which band-specific processing can apply large-support, channel-aware correction to the low band and local detail-preserving refinement to the high band. SandBAR therefore assigns generic mixing to the front FFN and reserves the rear FFN for band repair. The sandwich block is not simply a block with an additional FFN; its ordering implements the architectural principle of looking globally first and repairing by frequency afterwards.

For Split and the unsplit variant, post-attention placement changes mean PSNR by $+0.303$ and $+0.139$ dB with global attention, and $-0.011$ and $-0.158$ dB with windowed attention.

### 4.5. Architectural ablation

We next isolate the remaining constituent designs. Under the controlled 100-epoch setting of Tables 4 and 5, model (a) is the full SandBAR scaffold. Replacing its content-adaptive splitter with a fixed Gaussian split in model (b) degrades PSNR by $3.69\%$ $\downarrow$ and SSIM by $1.02\%$ $\downarrow$, on every domain. This is because a fixed ker-

Table 1. **Comparison on the 4,800-pair underwater restoration benchmark** (published results from the source papers [3, 27]).

| Method | EUVP | | LSUI | | UIEB | 3-domain |
|---|---|---|---|---|---|---|
| | PSNR↑ | SSIM↑ | PSNR↑ | SSIM↑ | PSNR↑ | mean PSNR↑ |
| ROP [19] | 15.34 | 0.714 | 17.38 | 0.806 | 18.48 | 17.07 |
| ADPCC [32] | 15.20 | 0.692 | 16.20 | 0.763 | – | – |
| WWPF [29] | 15.95 | 0.648 | 17.90 | 0.739 | – | – |
| UIE-WD [21] | 17.80 | 0.760 | 19.23 | 0.803 | – | – |
| TACL [20] | 20.99 | 0.782 | 22.97 | 0.828 | – | – |
| USUIR [9] | 21.94 | 0.810 | 23.75 | 0.860 | – | – |
| PUGAN [5] | 22.58 | 0.820 | 23.14 | 0.836 | 20.52 | 22.08 |
| SFGNet [31] | 22.68 | 0.585 | 22.71 | 0.653 | 19.57 | 21.65 |
| TUDA [26] | 23.73 | 0.843 | 25.52 | 0.878 | 22.72 | 23.99 |
| Semi-UIR [12] | 24.59 | 0.821 | 25.40 | 0.843 | 23.64 | 24.54 |
| GUPDM [22] | 24.79 | 0.847 | 25.33 | 0.877 | – | – |
| CLUIE-Net [18] | 24.85 | 0.844 | 23.57 | 0.864 | – | – |
| U-Transformer [24] | 24.99 | 0.829 | 25.15 | 0.838 | 20.75 | 23.63 |
| FUnIE-GAN [13] | 22.62 | 0.720 | 21.66 | 0.744 | 19.17 | 21.15 |
| URSCT [25] | 25.74 | 0.855 | 25.87 | 0.883 | – | – |
| Ucolor [17] | 23.72 | – | 21.30 | – | 18.37 | 21.13 |
| UIR-PolyKernel [10] | 26.42 | 0.866 | 26.55 | 0.888 | 23.52 | 25.50 |
| MAC-Lookup [27] | 26.85 | 0.891 | 27.78 | 0.931 | 23.84 | 26.16 |
| DEEP-SEA [3] | 29.27 | 0.905 | 28.90 | 0.897 | – | – |
| SandBAR w/o PaST | 29.50 | 0.907 | 28.52 | 0.919 | 22.34 | 26.79 |
| **SandBAR (ours)** | **29.89** | **0.910** | **28.92** | **0.922** | 22.77 | **27.19** |

Table 2. **PaST inference accounting.** The SPARK-retained row evaluates the trained SandBAR model before strip-and-fold, with the frequency scaffold kept at inference. After strip-and-fold, SandBAR recovers the permanent graph's parameter and FLOP counts and matched latency, while preserving its three-domain mean PSNR.

| Graph | Params↓ | GFLOPs↓ | Latency (ms)↓ | | 3-domain |
|---|---|---|---|---|---|
| | | @256 | @256 | @512 | mean PSNR↑ |
| SPARK retained | 28,586,687 | 146.86 | 27.62 | 81.52 | 27.20 |
| **Post-fold** | 24,088,479 | 132.15 | 21.53 | 61.89 | 27.19 |
| SandBAR w/o PaST | 24,088,479 | 132.15 | 21.44 | 61.91 | 26.79 |

Peak memory↓: 489.0 / 488.2 MB at $256^2$ for the strip-and-fold PaST graph and the scaffold-free baseline. Fold precision in Supplementary.

Table 3. **Effect of band decomposition.** The unsplit variant retains the two repair branches. Scores are equal-domain means over EUVP, LSUI and UIEB, evaluated from floating-point predictions after scaffold removal.

| Training scaffold | PSNR↑ | SSIM↑ |
|---|---|---|
| Without band decomposition | 27.191 | 0.91499 |
| **SandBAR** | **27.317** | **0.91748** |

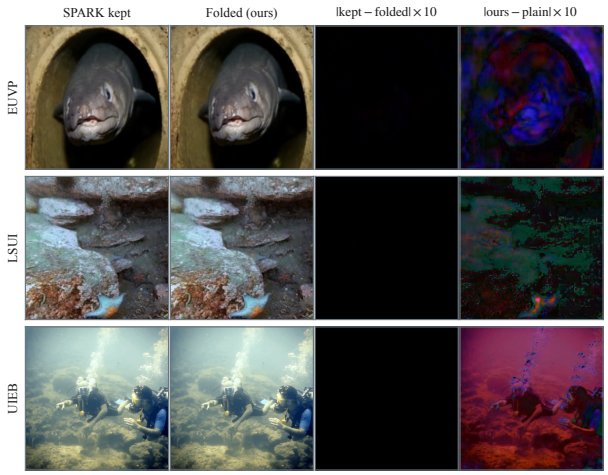



Figure 5. **Removing SPARK at inference is visually inert; training with it is not.** The first two columns evaluate the trained SandBAR model before and after strip-and-fold. The third column shows their absolute difference amplified $10\times$; the fourth applies the same amplification to the difference between the post-fold graph and the plain baseline. Depending on the domain, the mean absolute difference is 20–$139\times$ larger in the fourth column, so the near-black third column is a property of strip-and-fold rather than insufficient amplification.

nel cuts at a spatial frequency the image does not respect: it leaves the colour cast in the high band and structure in the low band, so the asymmetric operators are applied to mismatched content. Giving both bands the stronger low-band operator in model (c) costs $1.63\% \downarrow$ in PSNR, demonstrating the benefit of matching each operator to the dominant content of its band rather than enlarging both. Removing the low-frequency guidance path in model (d) costs $1.65\% \downarrow$—as much as the asymmetry itself—while the same path is worth $+0.02$ dB on the permanent graph alone, indicating

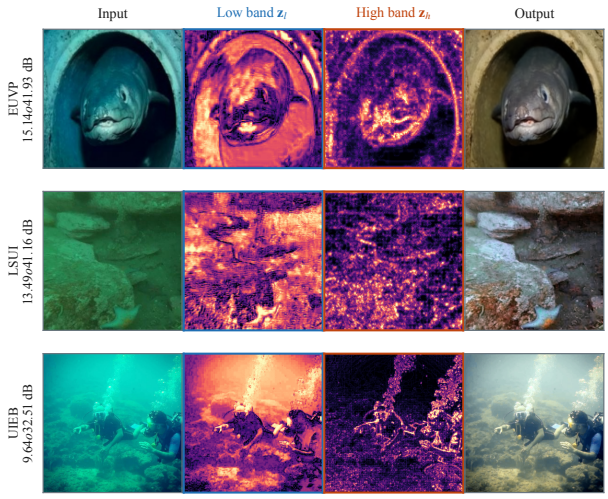



Figure 6. **What the band scaffold learns.** Channel-mean magnitudes of the SPARK band split ($\mathbf{z}_l$ and $\mathbf{z}_h$ columns; each map normalised independently) on one test image from each domain, captured from the *training* graph at the second encoder stage. The split is exact (Eq. (4)): the low branch carries smooth illumination and colour-cast structure, while the high branch isolates contours, bubble trails and rock texture. After strip-and-fold this machinery is absent at inference (Sec. 3.3); the plain graph that remains improves 464 of the 490 test images in both bands (Gaussian split, $\sigma$=10).

Table 4. **Where band repair belongs.** Three-domain mean PSNR under the controlled 100-epoch budget. Each row compares the same module in the pre-attention and post-attention feed-forward slots of one backbone, with all other settings held fixed. SPARK and SandBAR are ours, while DFFN and Restormer are existing designs.

| Configuration | | Mean PSNR↑ | |
|---|---|---|---|
| **Module** | **Backbone** | **pre-attn.** | **post-attn.** |
| SPARK | SandBAR | 26.43 | **26.72** |
| DFFN [14] | SandBAR | 26.25 | **26.52** |
| SPARK | Restormer [28] | 26.15 | **26.44** |

Table 5. **Architectural ablation.** Model (a) is the full SandBAR scaffold. All variants are trained under the same controlled 100-epoch budget as Table 4, with each variant changing exactly one design decision and all other settings held fixed. $\Delta$ is the three-domain mean PSNR relative to model (a).

| | Variant | Splitter | Band ops | LF guid. | PSNR↑ | $\Delta\uparrow$ |
|---|---|---|---|---|---|---|
| (a) | SandBAR full | adaptive | asym. | ✓ | **26.72** | — |
| (b) | fixed Gaussian | fixed | asym. | ✓ | 25.73 | −3.69% |
| (c) | symmetric ops | adaptive | sym. | ✓ | 26.29 | −1.63% |
| (d) | no LF guidance | adaptive | asym. | — | 26.28 | −1.65% |

that guidance pays off only where band-specific repair can act on what it delivers. These results confirm that SandBAR's three levels operate as one system across diverse underwater scenarios rather than as independent additions.

In the architectural ablations, floating-point evaluation gives 26.829 dB for the archived full model, 26.268 dB without FFT loss, and 26.586 dB with full-resolution supervision alone. The interaction between SPARK and LF guidance is +0.232 dB in mean PSNR without FFT supervision.

### 4.6. Qualitative results

Beyond the quantitative gains, SandBAR shows a clear visual advantage over the compared methods. As shown in Figure 4, separating each feature map into a low- and a high-frequency band and repairing the two with different operators lets the model correct the global colour shift without flattening the detail that survives the water column. The compared methods recover scene structure but not its colour: some carry the input's cast through to the output, others desaturate exactly the regions the reference renders most vividly. Our results keep the reference's colour distribution, preserve object boundaries and retain fine texture, which together indicate that band-asymmetric repair is what the visual improvement rests on.

## 5. Conclusion

Our main contribution is SandBAR, a novel unified band-asymmetric architecture for underwater image restoration. It turns the band-asymmetric tendency of large-scale colour distortion and local high-frequency detail loss into coordinated architectural targets across block topology, encoder guidance and three-scale spatial-frequency supervision. Within this architecture, the novel band-asymmetric sandwich block reserves a post-attention repair slot, and the novel SPARK module applies channel-aware large-support correction to the low band and lightweight local refinement to the high band, restoring global colour while preserving fine structure. To make this nonlinear band-repair capacity removable, PaST residual-anchors the scaffold and folds its remaining bias. The strengthened permanent SandBAR graph therefore runs at inference without the frequency scaffold, with unchanged parameters and FLOPs at matched latency. SandBAR establishes a new three-domain mean state of the art on the benchmark family, exceeding the strongest published result by approximately 1.0 dB. These results establish band asymmetry as an effective architectural prior for underwater restoration and removable nonlinear scaffolding as a promising route to stronger inference-efficient models. Extending removable nonlinear scaffolding across restoration tasks is an important direction for future work.

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